



University of Piraeus

School of Information and Telecommunications Technologies

Department of Digital Systems

Level: Postgraduate Program of Study

Network Security

Palo Alto

Supervisor: Prof. Christos Xenakis

Kalaitzidis Ioannis

ioannis\_kalaitzidis@ssl-unipi.gr

Piraeus

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## **Abstract**

Establish a secure network using Palo Alto Firewall in a VMware environment. Installation and configuration of the system, with the application of a strict security policy for selective access only to Web services, SSH and port 9001. A total ban on all other traffic through a "Deny-All" rule and confirmation of the sealing of the internal network through practical connectivity checks.

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# Introduction

The process begins with the download of the necessary .ova (Open Virtual Appliance) files, which are the pre-configured virtual machines that will make up the laboratory:

- Palo Alto Firewall
- Management Client
- DMZ Server
- Virtual Router

To understand the architecture of the lab, it is important to clarify the role of each virtual machine and the interaction between them in the context of network security. The Virtual Router acts as the connection node to the external network, simulating the Gateway to the internet (Outside Zone) and manages the routing of traffic to and from the outside world. The central pillar of the infrastructure is the Palo Alto Firewall, which intervenes between all network zones. Its role is to inspect and filter all data traffic, allowing or rejecting connections based on security policies.

Inside the network, the Client is placed in the Inside Zone and is used to manage the Firewall, having controlled access to the outside but remaining protected from incoming external threats. Instead, DMZ Server is placed in an isolated zone. Its purpose is to host services that need to be accessible from the internet (such as Web Server or SSH), while ensuring that, in the event of a breach, the attacker will not have direct access to the internal network (Client), as the traffic between DMZ and Inside Zone is strictly controlled by the Firewall.

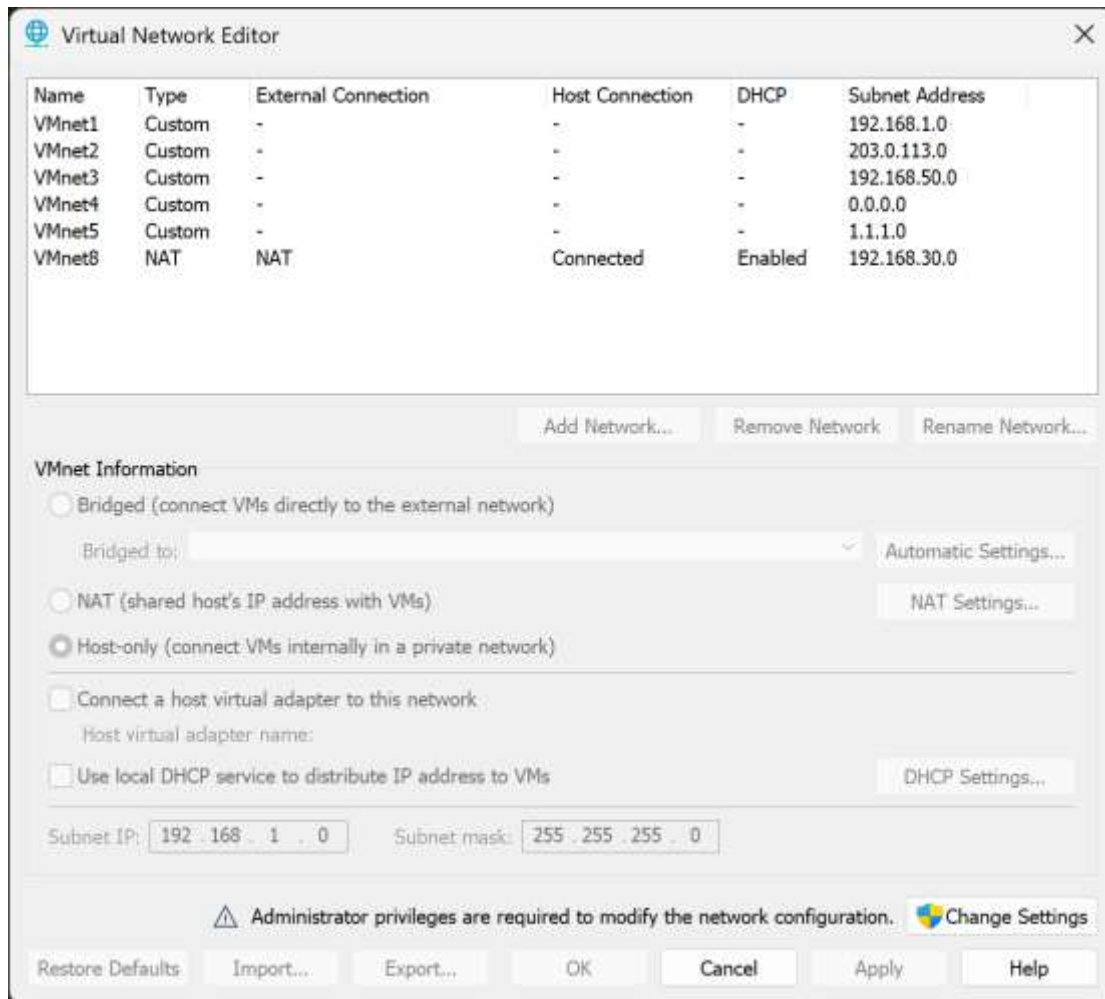
Before starting the machines, it is required to configure the network infrastructure in the VMware Workstation via the Virtual Network Editor. Specifically, virtual networks (VMnets) with the appropriate IP addresses (Subnets) are created and configured to ensure the correct routing of traffic and communication between zones (Inside, Outside, DMZ), according to the topology of the laboratory.

After the network is configured, the four virtual machines are imported into VMware. At this stage, a critical action is to manually set the MAC Addresses on the network cards of each machine, based on the driver's specifications, to avoid conflicts and correctly assign static IP addresses. Upon completion of the hardware

configurations, the machines are booted in a specific order (VRouter, Firewall, DMZ, Client) to ensure the smooth operation of the network services.

Access to the system is achieved through the Client virtual machine. From the Client environment, using a browser, we log in to the Firewall administration address (192.168.1.254) by entering the administrator credentials. As a final step of the preparation, the Authorization Code is entered to unlock the necessary functions and receive the updates. Now, the lab is fully operational and ready to implement security rules, such as enabling SSH and HTTP to DMZ and cutting off the rest of the traffic, as defined in the speech of the assignment.

## **Practical piece of work**

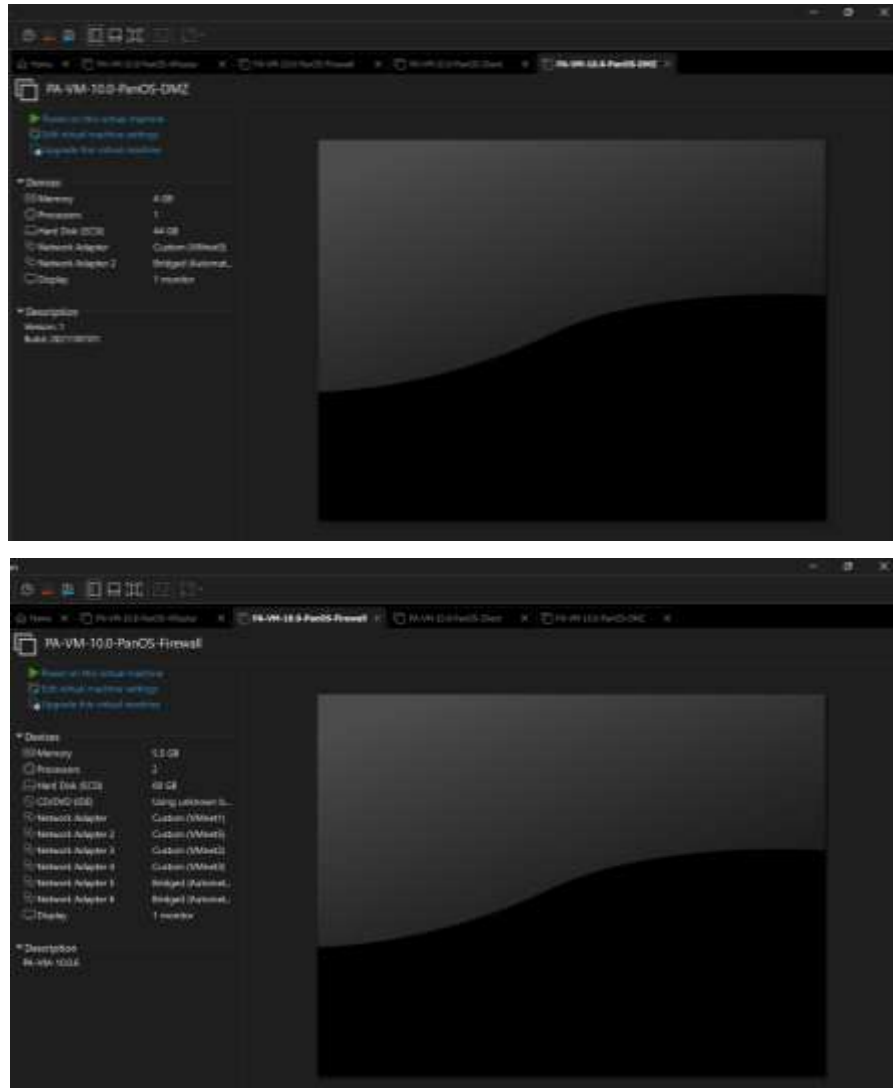


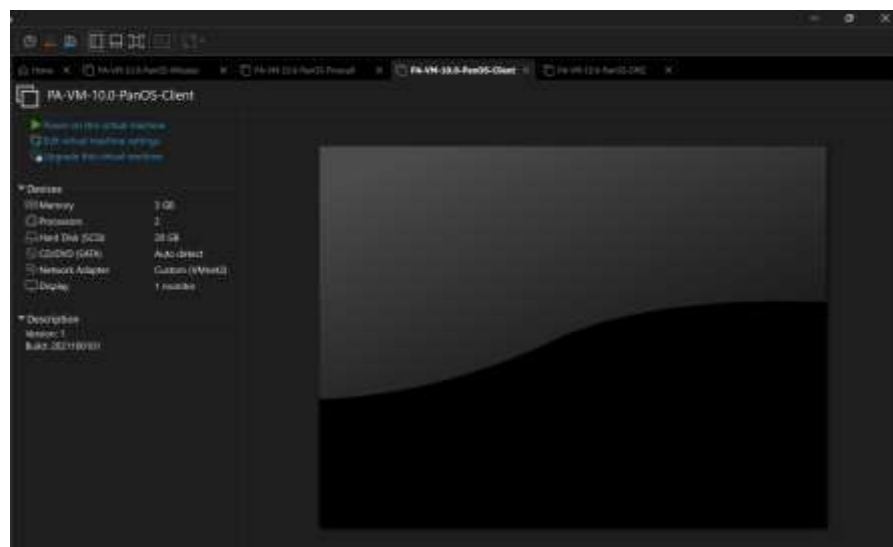
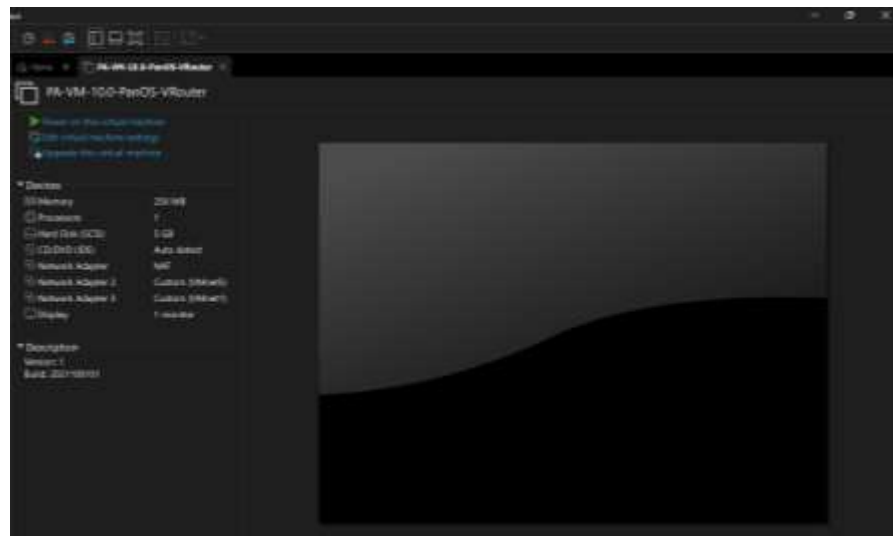
The implementation of the workshop began with the configuration of the Virtual Network Editor in VMware Workstation. Operating with administrative privileges, we proceeded to define the Subnet IP addresses, as well as to modify the MAC addresses of the network adapters, to keep up with the required topology. Then, we imported the virtual machines from the .ova files that were given to us, following the following starting order:

1. vRouter
2. Firewall (PA-VM)
3. DMZ Server
4. Client

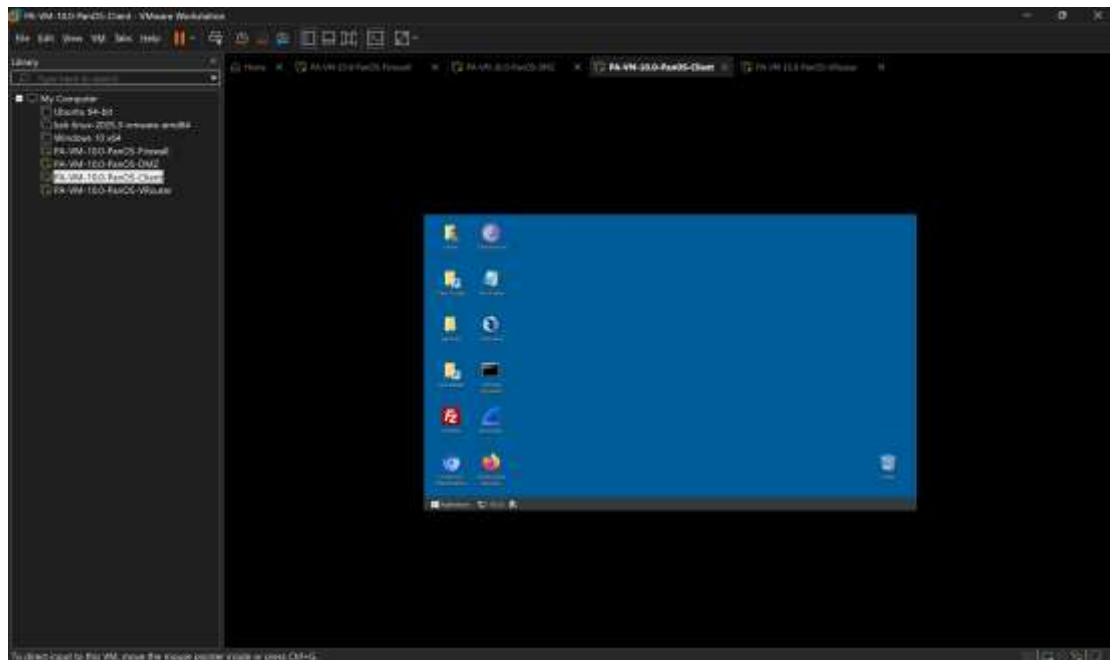
During the process we faced significant delay and technical problems, as there was a repeated loss of internet connectivity (network drop) on the computer (Host). The problem was found in the settings of the network cards, which were originally set to Bridged mode. This caused an IP Conflict, as the lab's static IPs (192.168.1.x) coincided with the address range of our local network, leading to a connection crash. The issue

was resolved by faithfully applying the instructions of the official Palo Alto Networks installation guide (Palo Alto Networks VMware Workstation 10.0 Academy Lab Deployment Guide). Specifically, we set up the adapters in Custom mode (VMnets) and NAT mode for the vRouter, ensuring the isolation of lab traffic from the physical network.

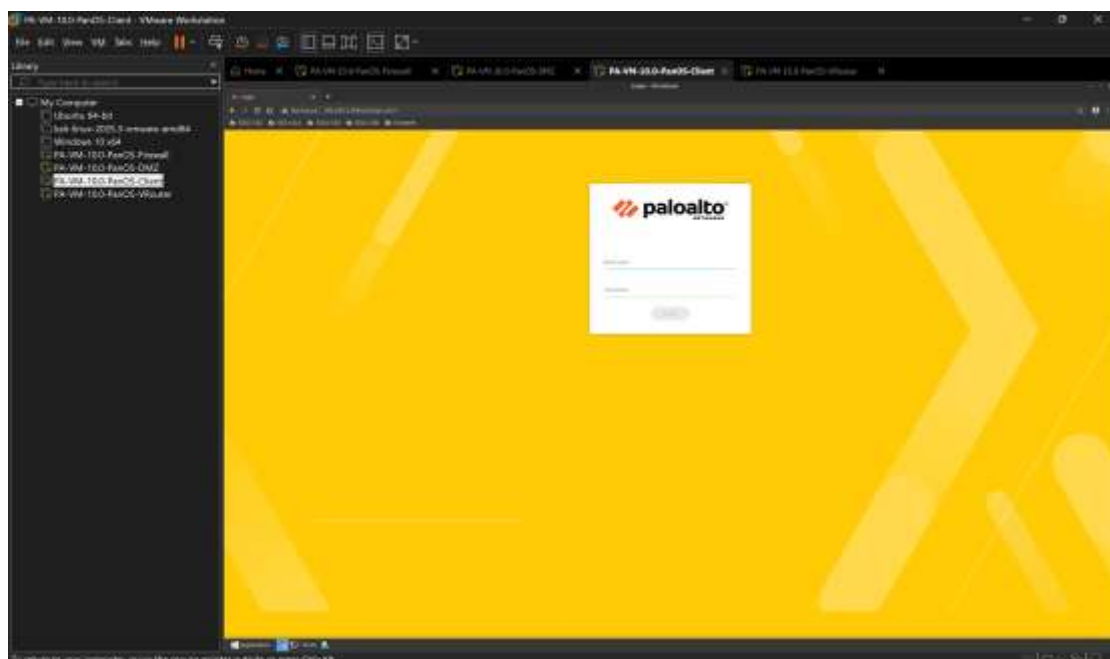




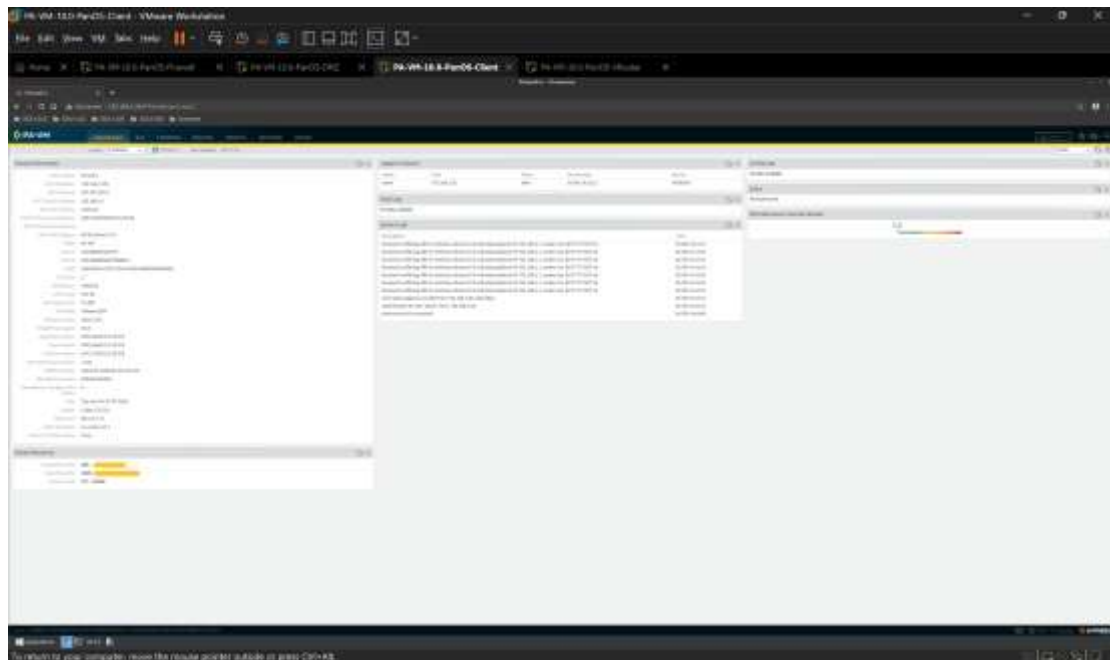
After applying the correct settings, the function was fully restored and the Client started successfully.



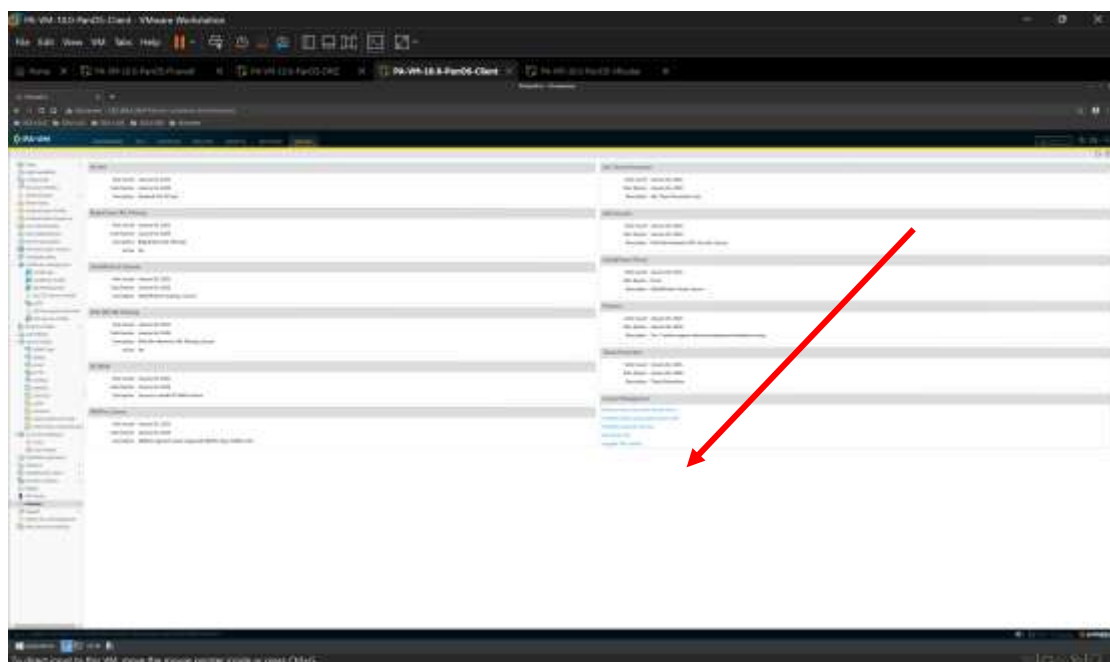
Following the steps of the lab guide, we logged into the Palo Alto Firewall management environment. Specifically, we opened the Chromium Web Browser in the Client and entered the address <https://192.168.1.254> to obtain access to the input console.



We enter the administrator codes and then the Palo Alto Firewall Web Interface is displayed.

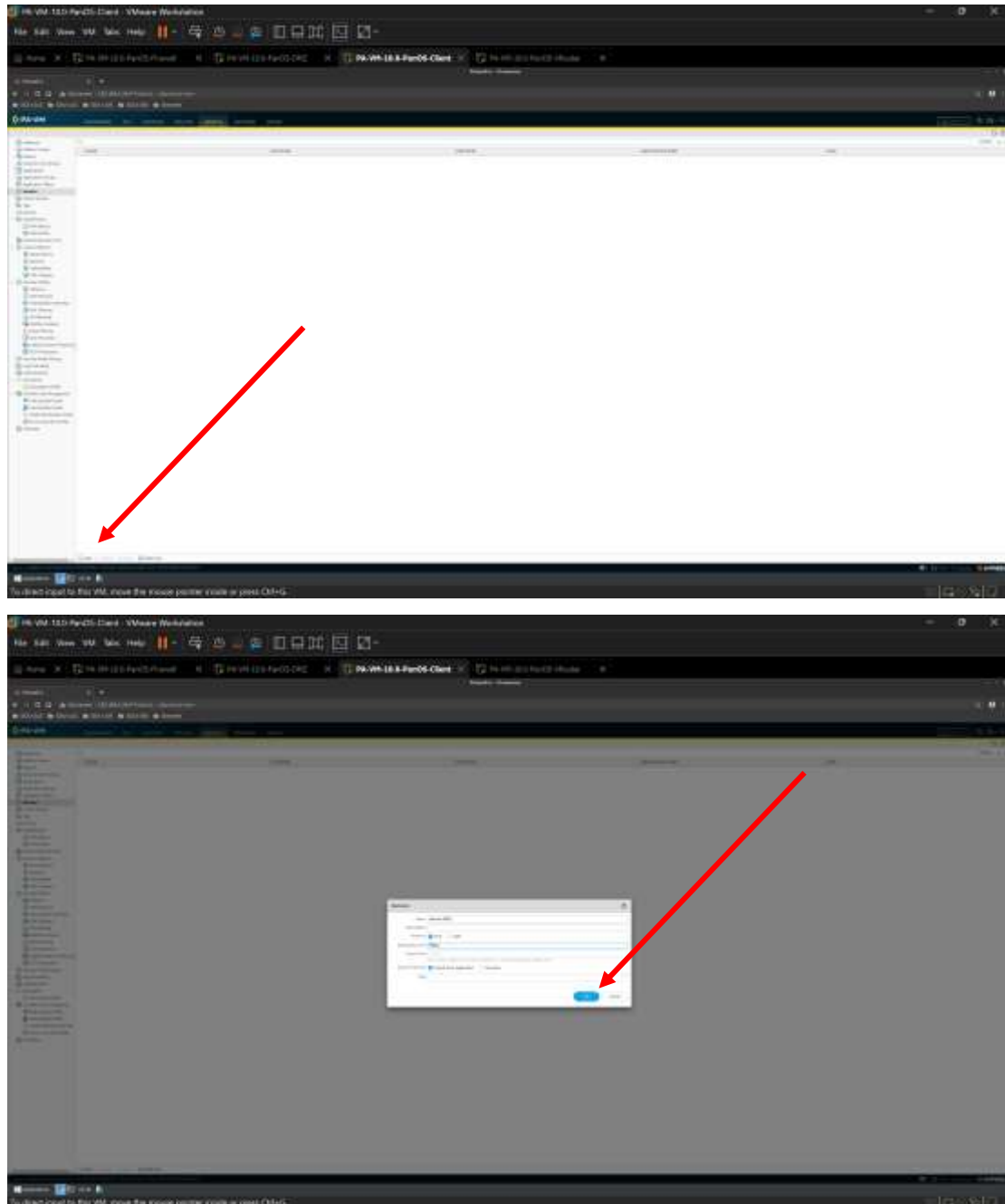


Faithfully following the procedure described in the installation guide, we completed the initial configuration. In the final stage, we went to the Device > Licenses tab and selected the "Activate features using authorization code" function. By entering the Authorization Code that was granted to us, we successfully activated the necessary software licenses, such as Threat Prevention and DNS Security, which is also confirmed in the screenshot below.

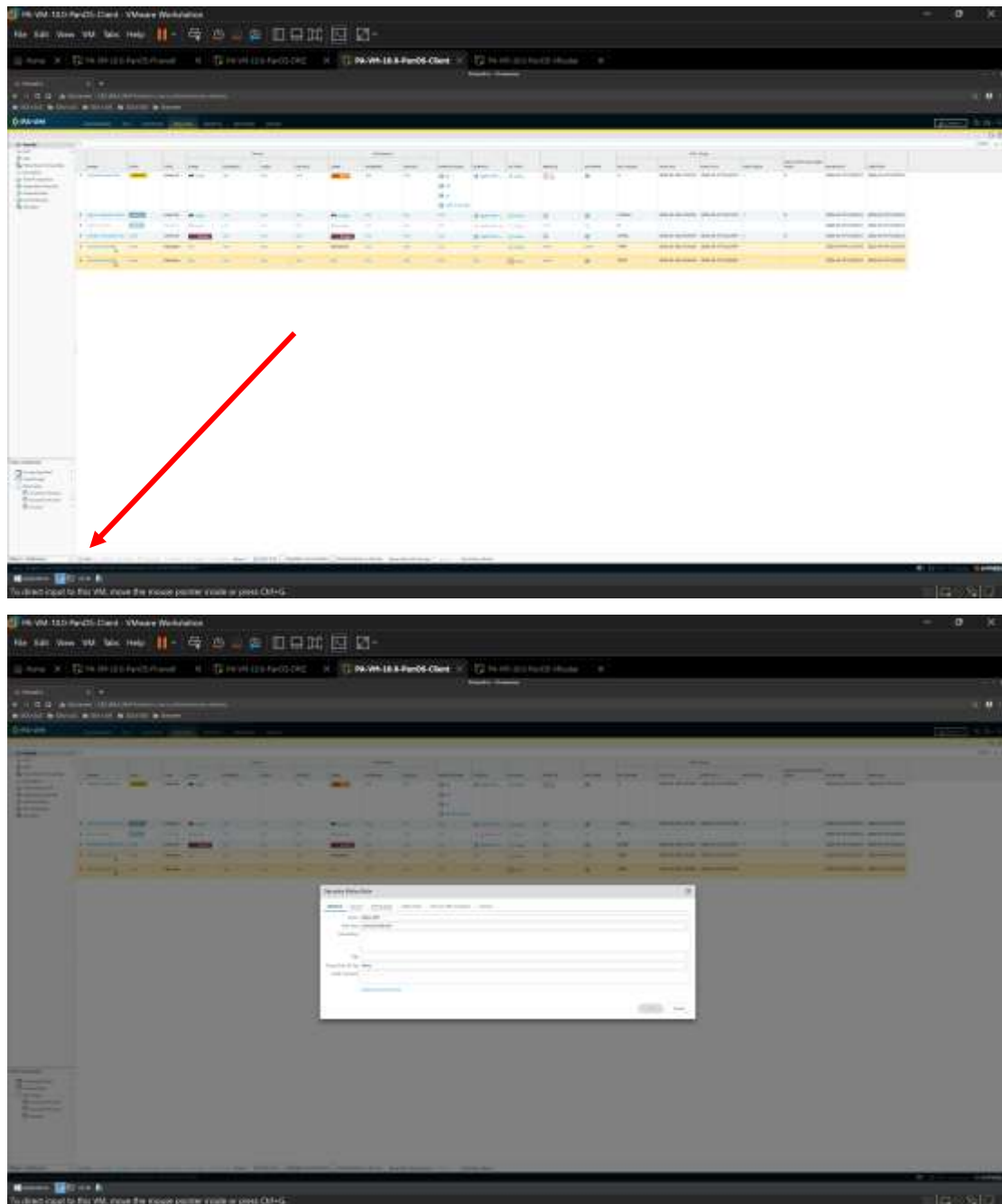


To meet the requirements of the Use Case, it was initially required to create a

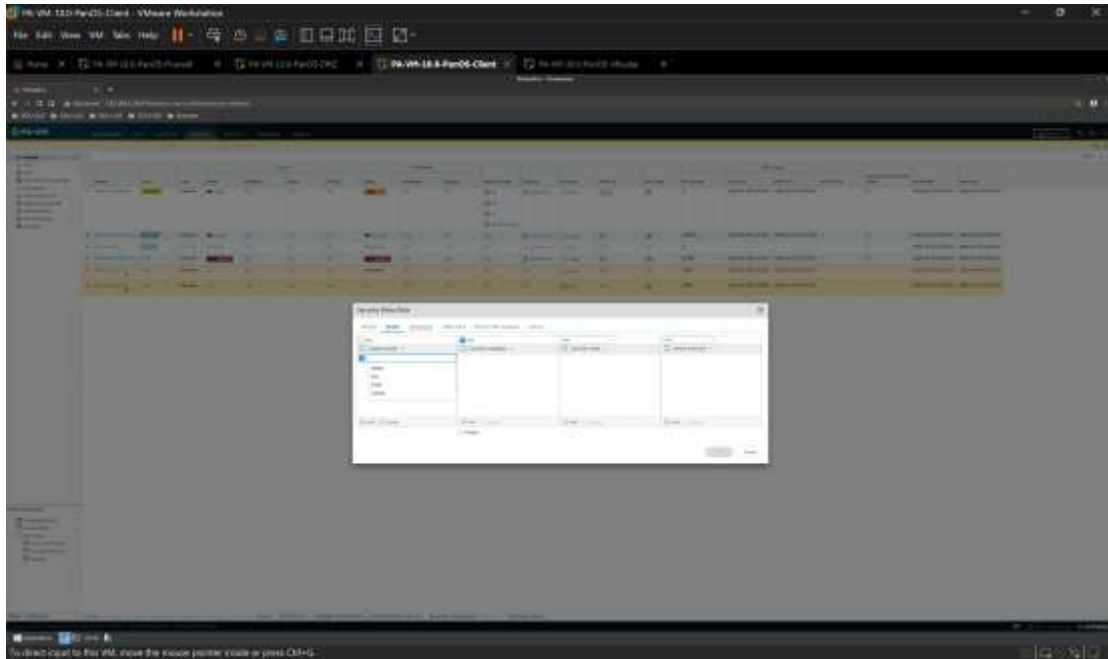
new Service Object, as port 9001 is not included in the default Firewall applications. Specifically, we went to the Objects > Services tab and created a new service called "Service-9001". TCP was defined as the communication protocol and 9001 as the Destination Port, as shown in the relevant screenshots.



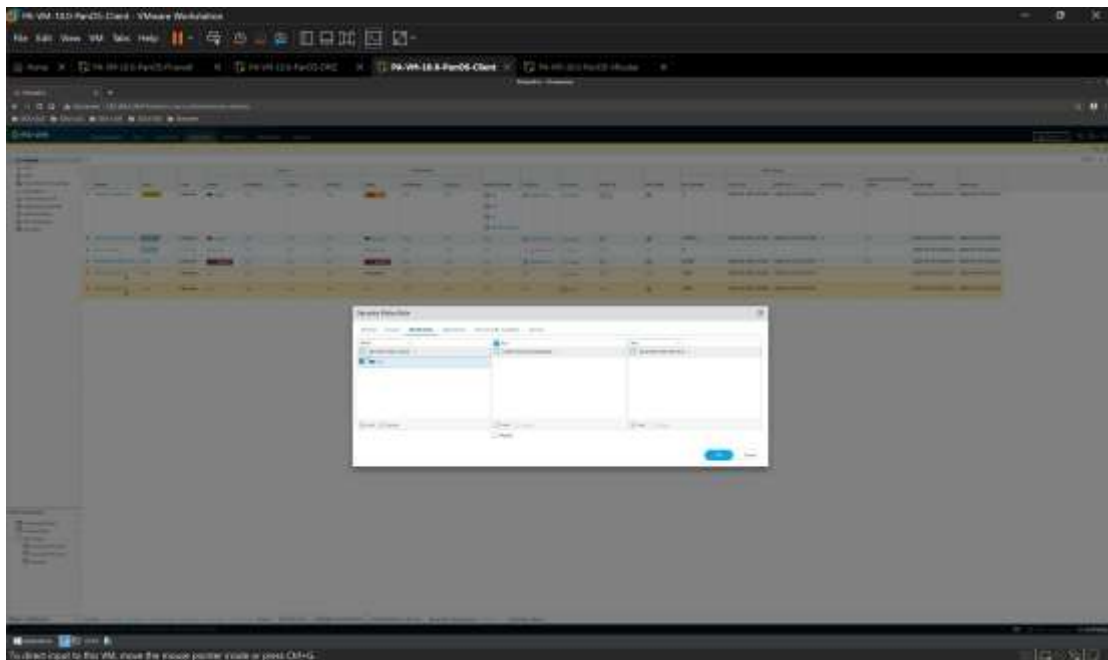
After creating the service object, we proceeded to the Policies > Security tab to implement the required rules. For the first rule called "Allow-SSH", we followed these step-by-step steps:



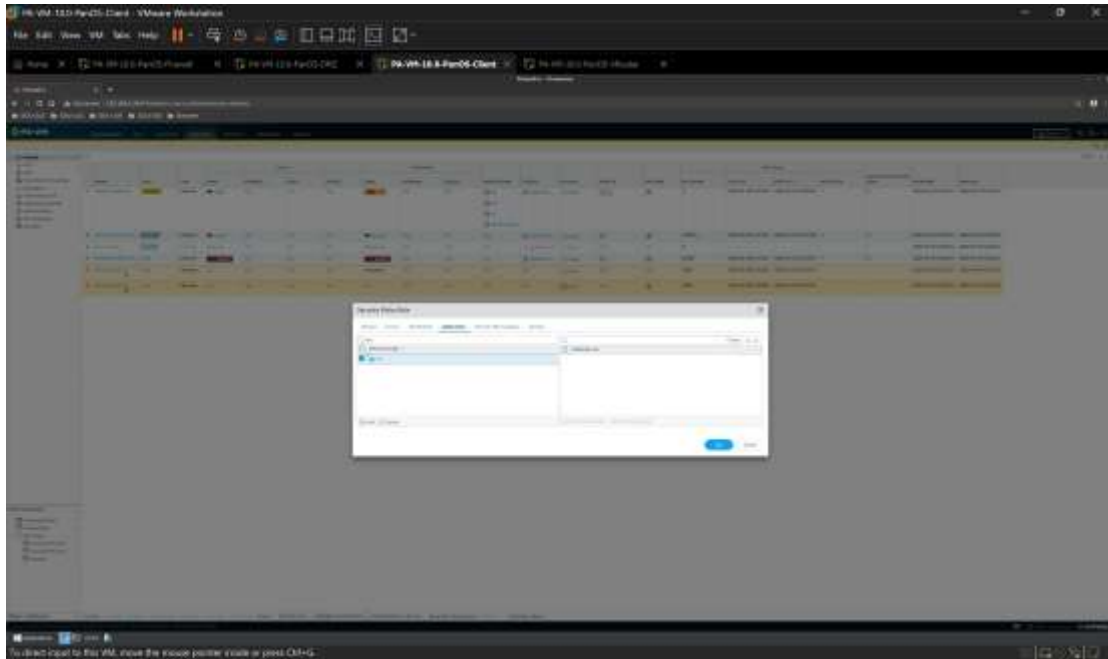
1. In the Source tab, we set the source zone to "outside", as the traffic is coming from the external network.



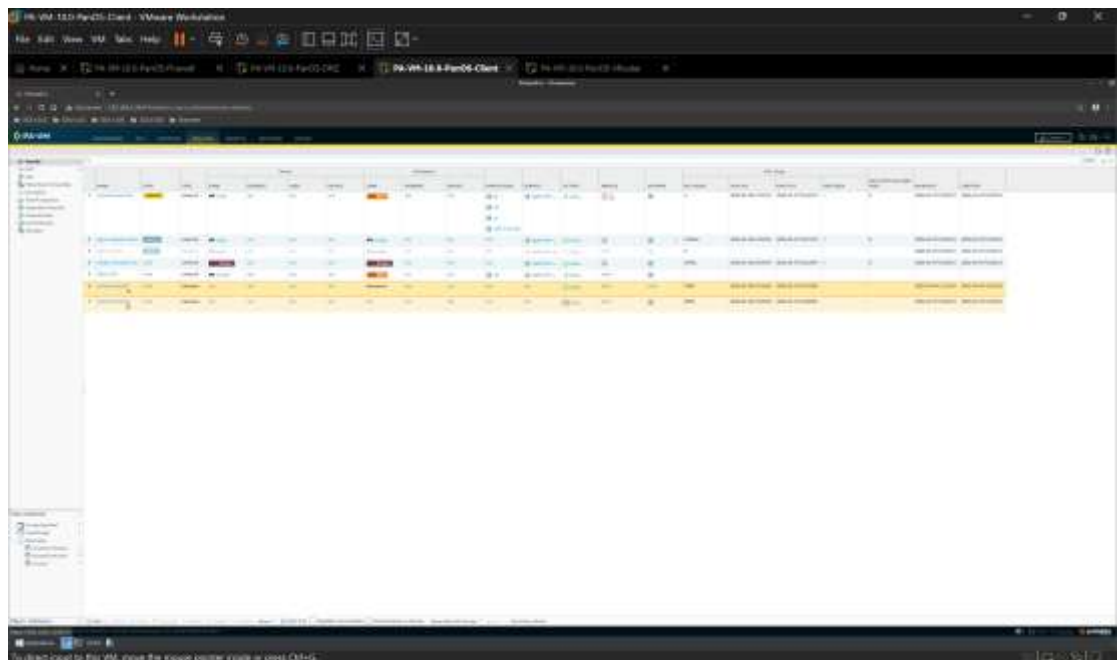
2. In the Destination tab, we selected the "dmz" zone, where our servers are hosted.



3. In the Application tab, we searched for and selected the 'ssh' app. This allows the Firewall to recognize the SSH protocol regardless of port.



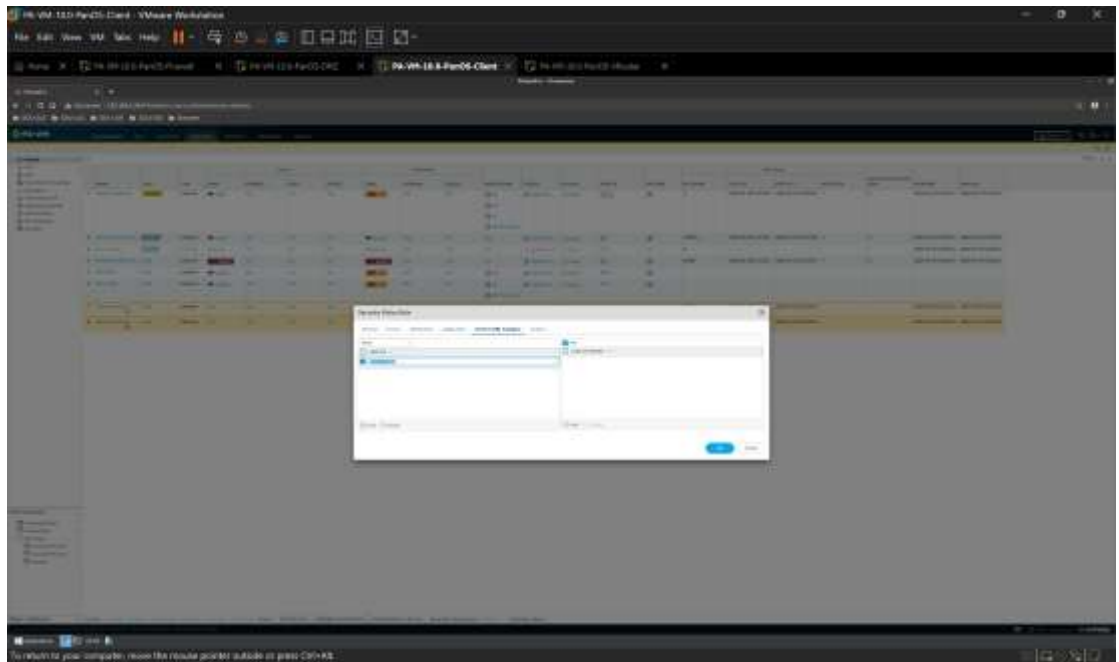
4. Finally, in the Actions tab, we made sure that the setting is set to "Allow". The following screenshot shows the complete list of Security Policy Rules that have been created. As can be seen, the order of the rules has been set correctly, with the allowed access (Allow) coming first and the Deny-All rule at the end, ensuring the hierarchical application of the policy.



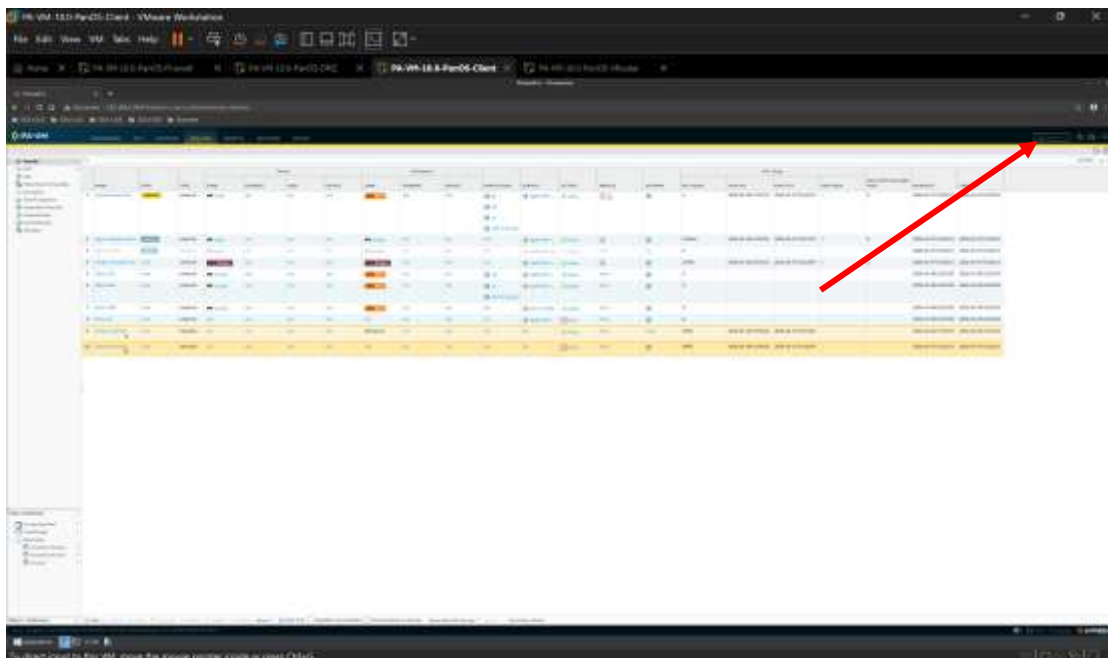
Following the same methodology, we created the rest of the rules of the exercise:

- **Allow-Web:** Defined for Source "outside" and Destination "dmz", selecting "web-browsing" and "ssl" in Applications to support traffic HTTP/HTTPS.

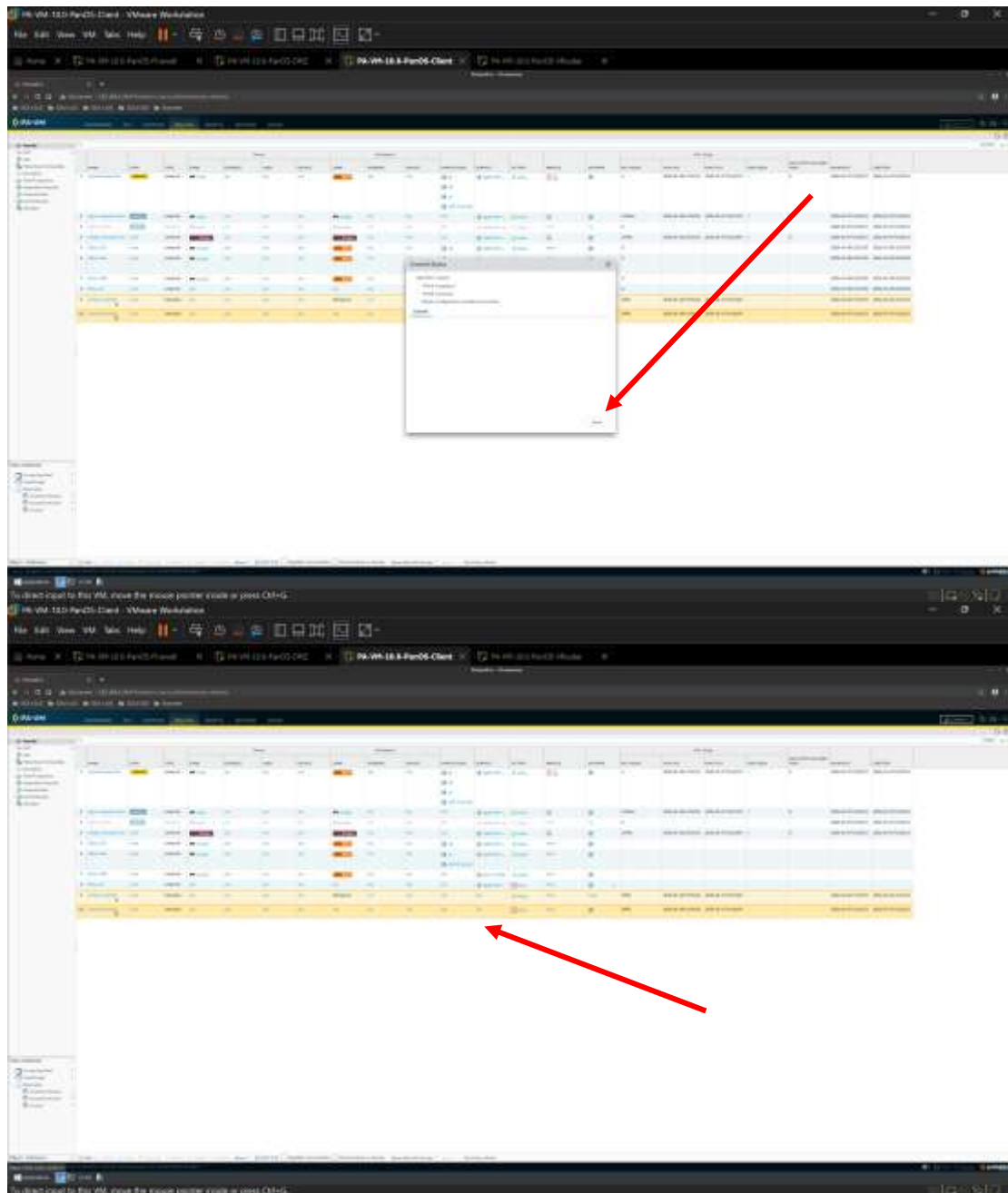
- **Allow-9001:** Applied for Source "outside" and Destination "dmz". In this rule, instead of Application, we selected the "Service-9001" object that we had previously created in the Service tab.



- **Deny-All:** As the last safety rule, a deny-all rule has been added for any zone (Any) to any zone (Any). This rule is critical as it ensures that any movement not explicitly allowed by the previous rules will be rejected.

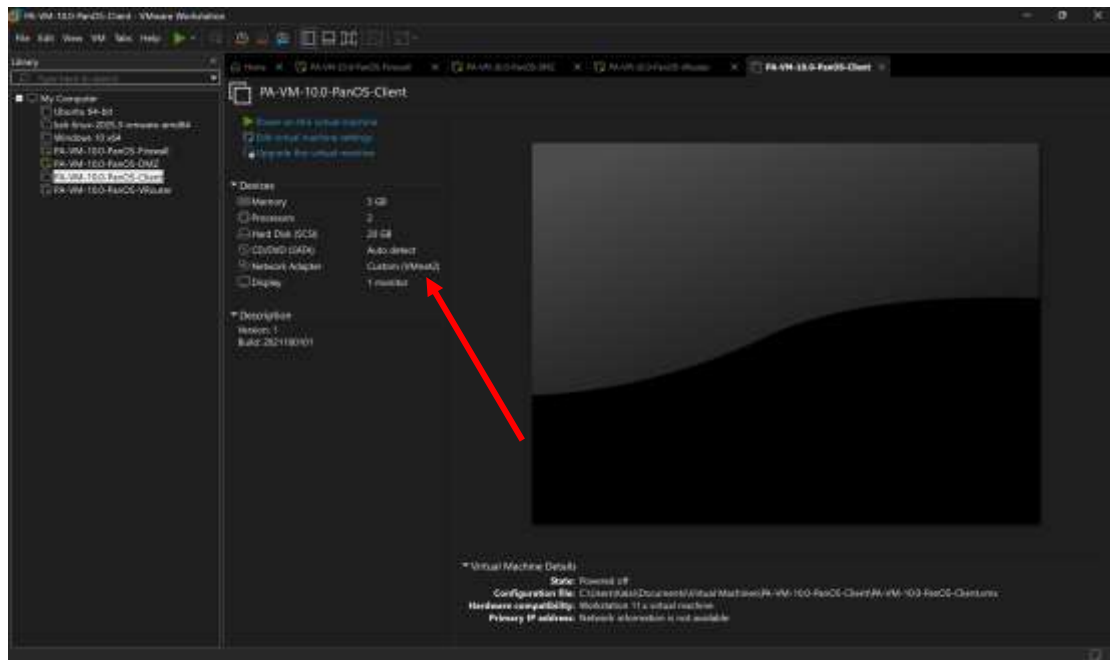


With the successful entry of all four security rules, we proceeded to finalize the settings by selecting Commit. The system confirmed the implementation of the policy ("Configuration committed successfully").

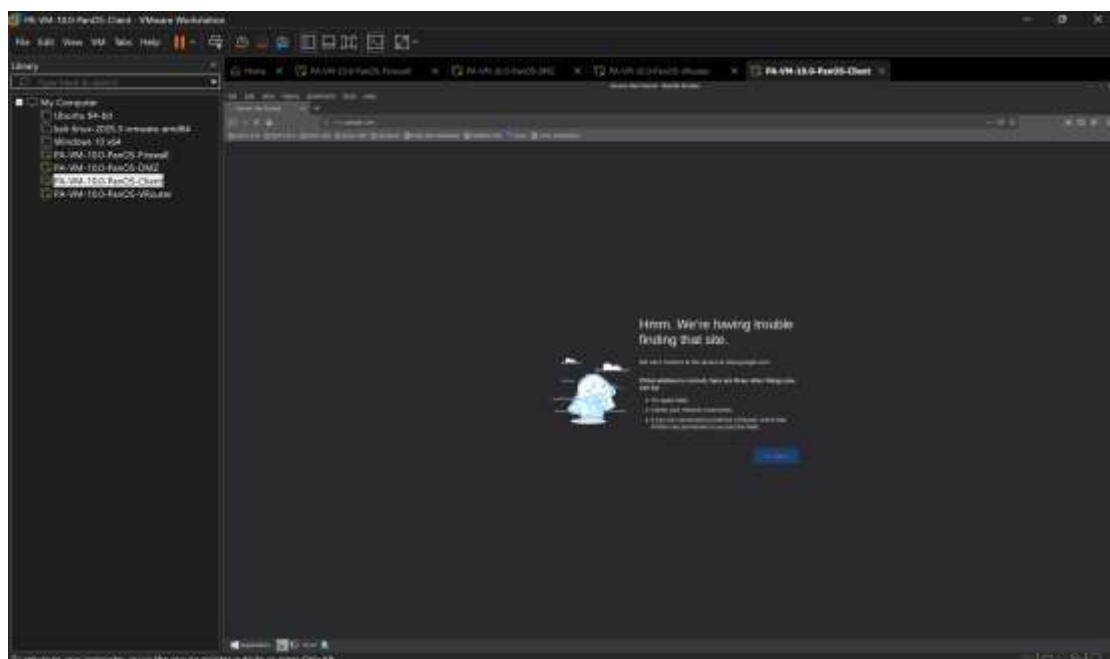


To practically verify the safety and proper functioning of the rules, we carried out the following control steps:

1. **Change Client Network:** First, we changed the settings of the Client virtual machine, moving the Network Adapter from the management network (VMnet1) to the internal Custom network (VMnet2). In this way, the Client now simulates a simple user behind the Firewall and not the system administrator.

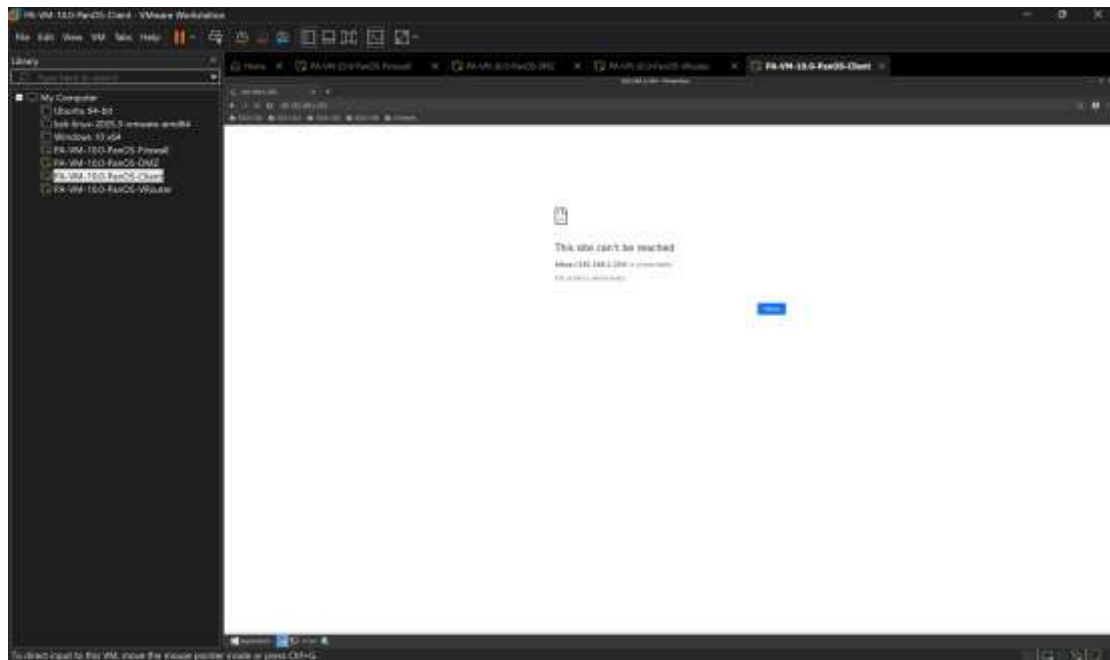


2. **Internet Access Control (Deny-All):** We tried to connect to a random website (e.g. Google) via the browser. As shown in the relevant screenshot, the connection failed with the message "Server Not Found". This result is expected and desirable. It confirms that the last "Deny-All" rule works correctly, rejecting any traffic (such as Internet access) that has not been explicitly allowed in the previous rules.



3. **Management Security Check:** Next, we tried to regain access to the Firewall administration page (192.168.1.254). The connection failed with the message "This site can't be reached". This confirms the proper isolation of the networks.

The Client, now in the Inside Zone, does not have and should not access the Management Interface, thus protecting the Firewall from unauthorized interference.



In conclusion, the lab was successfully completed, as the Firewall only allows the specified services (SSH, Web, Port 9001 to DMZ) and effectively blocks any other unauthorized traffic.

## Bibliography

Shesh Chauhan IT Trainer. (2025, December 22). *Palo Alto Firewall Lab Guide | Step-by-Step Configuration Tutorial* | [Video file]. Retrieved from <https://www.youtube.com/watch?v=K2S7B4XDEXQ>

Palo Alto Networks. (2021). *Palo Alto Networks VMware Workstation 10.0 Academy Labs deployment guide* [PDF document].